Class Activity 3rd Aug 2026

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Course Code: CSA0903
Course Name: Programming in JAVA

1. ArithmeticException – Student Calculator

Problem Statement:

Develop a Java program to calculate the average marks of students. The program should accept the **total marks** and the **number of subjects** from the user. If the number of subjects entered is **0**, the program should handle the exception gracefully and display an appropriate error message instead of terminating abruptly.

Requirements:

- Read total marks and number of subjects.
- Calculate and display the average.
- Handle `ArithmeticException` using `try-catch`.
- Display a user-friendly error message when division by zero occurs.

Code:

StudentCalculator.java | Share | Run

```java
import java.util.Scanner;

public class StudentCalculator {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        try {
            System.out.print("Enter Total Marks: \n");
            int total = sc.nextInt();

            System.out.print("Enter Number of Subjects: \n");
            int subjects = sc.nextInt();

            int average = total / subjects;

            System.out.println("Average Marks = " + average);
        } catch (ArithmeticException e) {
            System.out.println("Error: Number of subjects cannot be zero.");
        }

        sc.close();
    }
}
```

Output

```
Status: Accepted
Memory Used: 13136 byte | Execution Time: 0.219 sec

Enter Total Marks:
Enter Number of Subjects:
Average Marks = 133
```

Input

```
400
3
```

2. NullPointerException – Employee Management System

Problem Statement:

Develop a Java program for an Employee Management System. The program should create an employee object and display the employee's name. Demonstrate a situation where the employee object is **not initialized (null)** and handle the resulting exception gracefully.

Requirements:

- Create an `Employee` class with employee ID and name.
- Attempt to access the employee object's data without creating the object.
- Handle `NullPointerException` using `try-catch`.
- Display an appropriate message instead of allowing the program to crash.

Code:

```java
class Employee {
    int id;
    String name;

    Employee(int id, String name) {
        this.id = id;
        this.name = name;
    }
}

public class EmployeeManagement {
    public static void main(String[] args) {

        Employee emp = null;

        try {
            System.out.println("Employee Name: " + emp.name);
        } catch (NullPointerException e) {
            System.out.println("Error: Employee object is not initialized.");
        }
    }
}
```

Status: Accepted

Memory Used: 9952 byte | Execution Time: 0.085 sec

```
Error: Employee object is not initialized.
```

Input

12
HELLO WORLD

3. ArrayIndexOutOfBoundsException – Student Marks Processing

Problem Statement:

Develop a Java program to store the marks of **five students** in an array. Allow the user to enter the index of the student whose mark should be displayed. If the entered index is outside the valid range, handle the exception and display an informative message.

Requirements:

- Store marks in an integer array of size 5.
- Accept an index from the user.
- Display the mark corresponding to the entered index.
- Handle `ArrayIndexOutOfBoundsException` using `try-catch`.
- Inform the user about the valid index range.

Code:

StudentMarks.java | Share | Run | Output

```java
import java.util.Scanner;

public class StudentMarks {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int marks[] = {85, 90, 75, 88, 95};

        try {
            System.out.print("Enter Student Index (0-4): ");
            int index = sc.nextInt();

            System.out.println("Marks = " + marks[index]);
        } catch (ArrayIndexOutOfBoundsException e) {
            System.out.println("Error: Invalid index. Enter a value between 0 and 4.");
        }

        sc.close();
    }
}
```

```
Status: Accepted
Memory Used: 12332 byte | Execution Time: 0.185 sec

Enter Student Index (0-4): Error: Invalid index. Enter a value between 0 and 4.
```

Input

```
5
```

4. NumberFormatException – Customer Registration System

Problem Statement:

Develop a Java program for a Customer Registration System. The program should accept the customer's **age as a String** and convert it into an integer. If the user enters invalid data (such as alphabets or special characters), the program should handle the exception and display an appropriate error message.

Requirements:

- Read age as a String.
- Convert the String into an integer using `Integer.parseInt()`.
- Display the customer's age if the input is valid.
- Handle `NumberFormatException` using `try-catch`.
- Display a meaningful message when invalid input is entered.

Code:

```java
import java.util.Scanner;

public class CustomerRegistration {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        try {
            System.out.print("Enter Age: ");
            String age = sc.nextLine();

            int customerAge = Integer.parseInt(age);

            System.out.println("Customer Age = " + customerAge);
        } catch (NumberFormatException e) {
            System.out.println("Error: Please enter a valid numeric age.");
        }

        sc.close();
    }
}
```

```
Status: Accepted
Memory Used: 12316 byte | Execution Time: 0.169 sec

Enter Age: Error: Please enter a valid numeric age.
```

Input

```
six
```